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Code Snippets

point.h

```
struct Point;
struct Point* makePoint(double x, double y);
double distance (struct Point *p1, struct Point *p2);
```

point.c

```
#include "point.h"
#include <stdlib.h>
#include <math.h>

struct Point {
  double x,y;
};

struct Point* makepoint(double x, double y) {
  struct Point* p = malloc(sizeof(struct Point));
  p->x = x;
  p->y = y;
  return p;
}

double distance(struct Point* p1, struct Point* p2) {
  double dx = p1->x - p2->x;
  double dy = p1->y - p2->y;
  return sqrt(dx*dx+dy*dy);
}
```

point.h

```
class Point {
public:
  Point(double x, double y);
  double distance(const Point& p) const;

private:
  double x;
  double y;
};
```

point.cc

```
#include "point.h"
#include <math.h>

Point::Point(double x, double y)
: x(x), y(y)
{}

double Point::distance(const Point& p) const {
  double dx = x-p.x;
  double dy = y-p.y;
  return sqrt(dx*dx + dy*dy);
}
```

namedPoint.h

```
struct NamedPoint;

struct NamedPoint* makeNamedPoint(double x, double y, char* name);
void setName(struct NamedPoint* np, char* name);
char* getName(struct NamedPoint* np);
```

namedPoint.c

```
#include "namedPoint.h"
#include <stdlib.h>

struct NamedPoint {
  double x,y;
  char* name;
};

struct NamedPoint* makeNamedPoint(double x, double y, char* name) {
  struct NamedPoint* p = malloc(sizeof(struct NamedPoint));
  p->x = x;
  p->y = y;
  p->name = name;
  return p;
}

void setName(struct NamedPoint* np, char* name) {
  np->name = name;
}

char* getName(struct NamedPoint* np) {
  return np->name;
}
```

main.c

```
#include "point.h"
#include "namedPoint.h"
#include <stdio.h>

int main(int ac, char** av) {
  struct NamedPoint* origin = makeNamedPoint(0.0, 0.0, "origin");
  struct NamedPoint* upperRight = makeNamedPoint
    (1.0, 1.0, "upperRight");
  printf("distance=%f\n",
    distance(
              (struct Point*) origin,
              (struct Point*) upperRight));
}
```

```
#include <stdio.h>

void copy() {
   int c;
   while ((c=getchar()) != EOF)
      putchar(c);
}
```

```
struct FILE {
  void (*open)(char* name, int mode);
  void (*close)();
  int (*read)();
  void (*write)(char);
  void (*seek)(long index, int mode);
};
```

```
#include "file.h"

void open(char* name, int mode) {/*...*/}
void close() {/*...*/};
int read() {int c;/*...*/ return c;}
void write(char c) {/*...*/}
void seek(long index, int mode) {/*...*/}

struct FILE console = {open, close, read, write, seek};
```

```
extern struct FILE* STDIN;


int getchar() {
   return STDIN->read();
}
```